# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 24 | Height splits the base: $h^2 = 30^2 - 18^2 = 576$ |
| 2 | **(d)** 84 | Other side $= 24$; perimeter $= 2(18 + 24)$ |
| 3 | **(b)** 48 | Half of other diagonal $= \sqrt{30^2 - 18^2} = 24$ |
| 4 | **(d)** 30 | Height difference 18 m; $\sqrt{18^2 + 24^2} = 30$ |
| 5 | **(b)** 216 | Other leg $= 24$; area $= \frac{1}{2}ab$ |
| 6 | **(a)** 33, 44, 55 | Multiples of $(3, 4, 5)$ by 11 stay a triplet |
| 7 | **(c)** 36 | Height splits the base: $h^2 = 39^2 - 15^2 = 1296$ |
| 8 | **(b)** 102 | Other side $= 36$; perimeter $= 2(15 + 36)$ |
| 9 | **(d)** 72 | Half of other diagonal $= \sqrt{39^2 - 15^2} = 36$ |
| 10 | **(d)** 39 | Height difference 15 m; $\sqrt{15^2 + 36^2} = 39$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Height bisects the base, $h^2 = 30^2 - 18^2$ (1); $h = 24$ cm (1). | 1+1 |
| 12 | Other side $= 32$ cm (1); area $= 768$ cm² (1). | 1+1 |
| 13 | Half of other diagonal$^2 = 50^2 - 30^2 = 1600$ (1); other diagonal $= 80$ cm (1). | 1+1 |
| 14 | Height bisects the base, $h^2 = 39^2 - 15^2$ (1); $h = 36$ cm (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Initial foot distance $= \sqrt{25^2 - 24^2} = 7$ m (1); new height $= \sqrt{25^2 - 15^2} = 20$ m (1); slip $= 24 - 20 = 4$ m (1). | 1+1+1 |
| 16 | $18^2 + 24^2 = 900 = 30^2$ (1); right-angled with legs 18, 24 (1); area $= \frac{1}{2} \times 18 \times 24 = 216$ cm² (1). | 1+1+1 |
| 17 | Hypotenuse $= 30$ cm (1); area $= \frac{1}{2} \times 18 \times 24 = 216$ (1); $\frac{1}{2} \times 30 \times h = 216$ gives $h = 14.4$ cm (1). | 1+1+1 |
| | **OR:** Initial foot distance $= \sqrt{25^2 - 20^2} = 15$ m (1); new height $= \sqrt{25^2 - 20^2} = 15$ m (1); slip $= 20 - 15 = 5$ m (1). | 1+1+1 |
| 18 | $15^2 + 36^2 = 1521 = 39^2$ (1); right-angled with legs 15, 36 (1); area $= \frac{1}{2} \times 15 \times 36 = 270$ cm² (1). | 1+1+1 |
| | **OR:** Hypotenuse $= 39$ cm (1); area $= \frac{1}{2} \times 15 \times 36 = 270$ (1); $\frac{1}{2} \times 39 \times h = 270$ gives $h = 13.85$ cm (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) $\sqrt{18^2 + 24^2} = 30$ m (2); (ii) saved $= 18 + 24 - 30 = 12$ m (1); (iii) total $= 18 + 24 + 30 = 72$ m (2). | 2+1+2 |
| | **OR:** Height$^2 = 40^2 - 24^2 = 1024$, height $= 32$ cm (2); area $= \frac{1}{2} \times 48 \times 32 = 768$ cm² (2); perimeter $= 2 \times 40 + 48 = 128$ cm (1). | 2+2+1 |
| 20 | (i) $\sqrt{15^2 + 36^2} = 39$ m (2); (ii) saved $= 15 + 36 - 39 = 12$ m (1); (iii) total $= 15 + 36 + 39 = 90$ m (2). | 2+1+2 |
| | **OR:** Height$^2 = 35^2 - 21^2 = 784$, height $= 28$ cm (2); area $= \frac{1}{2} \times 42 \times 28 = 588$ cm² (2); perimeter $= 2 \times 35 + 42 = 112$ cm (1). | 2+2+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

## ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 270 | Other leg $= 36$; area $= \frac{1}{2}ab$ |
| 2 | **(d)** 35, 84, 91 | Multiples of $(5, 12, 13)$ by 7 stay a triplet |
| 3 | **(b)** 32 | Height splits the base: $h^2 = 40^2 - 24^2 = 1024$ |
| 4 | **(c)** 112 | Other side $= 32$; perimeter $= 2(24 + 32)$ |
| 5 | **(a)** 64 | Half of other diagonal $= \sqrt{40^2 - 24^2} = 32$ |
| 6 | **(d)** 40 | Height difference 24 m; $\sqrt{24^2 + 32^2} = 40$ |
| 7 | **(a)** 384 | Other leg $= 32$; area $= \frac{1}{2}ab$ |
| 8 | **(a)** 40, 75, 85 | Multiples of $(8, 15, 17)$ by 5 stay a triplet |
| 9 | **(d)** 28 | Height splits the base: $h^2 = 35^2 - 21^2 = 784$ |
| 10 | **(a)** 98 | Other side $= 28$; perimeter $= 2(21 + 28)$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Other side $= 28$ cm (1); area $= 588$ cm² (1). | 1+1 |
| 12 | Half of other diagonal$^2 = 50^2 - 14^2 = 2304$ (1); other diagonal $= 96$ cm (1). | 1+1 |
| 13 | Height bisects the base, $h^2 = 40^2 - 24^2$ (1); $h = 32$ cm (1). | 1+1 |
| 14 | Other side $= 40$ cm (1); area $= 1200$ cm² (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Initial foot distance $= \sqrt{13^2 - 12^2} = 5$ m (1); new height $= \sqrt{13^2 - 12^2} = 5$ m (1); slip $= 12 - 5 = 7$ m (1). | 1+1+1 |
| 16 | $24^2 + 32^2 = 1600 = 40^2$ (1); right-angled with legs 24, 32 (1); area $= \frac{1}{2} \times 24 \times 32 = 384$ cm² (1). | 1+1+1 |
| 17 | Hypotenuse $= 40$ cm (1); area $= \frac{1}{2} \times 24 \times 32 = 384$ (1); $\frac{1}{2} \times 40 \times h = 384$ gives $h = 19.2$ cm (1). | 1+1+1 |
| | **OR:** Initial foot distance $= \sqrt{17^2 - 15^2} = 8$ m (1); new height $= \sqrt{17^2 - 15^2} = 8$ m (1); slip $= 15 - 8 = 7$ m (1). | 1+1+1 |
| 18 | $21^2 + 28^2 = 1225 = 35^2$ (1); right-angled with legs 21, 28 (1); area $= \frac{1}{2} \times 21 \times 28 = 294$ cm² (1). | 1+1+1 |
| | **OR:** Hypotenuse $= 35$ cm (1); area $= \frac{1}{2} \times 21 \times 28 = 294$ (1); $\frac{1}{2} \times 35 \times h = 294$ gives $h = 16.8$ cm (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) $\sqrt{24^2 + 32^2} = 40$ m (2); (ii) saved $= 24 + 32 - 40 = 16$ m (1); (iii) total $= 24 + 32 + 40 = 96$ m (2). | 2+1+2 |
| | **OR:** Height$^2 = 50^2 - 30^2 = 1600$, height $= 40$ cm (2); area $= \frac{1}{2} \times 60 \times 40 = 1200$ cm² (2); perimeter $= 2 \times 50 + 60 = 160$ cm (1). | 2+2+1 |
| 20 | (i) $\sqrt{21^2 + 28^2} = 35$ m (2); (ii) saved $= 21 + 28 - 35 = 14$ m (1); (iii) total $= 21 + 28 + 35 = 84$ m (2). | 2+1+2 |
| | **OR:** Height$^2 = 50^2 - 14^2 = 2304$, height $= 48$ cm (2); area $= \frac{1}{2} \times 28 \times 48 = 672$ cm² (2); perimeter $= 2 \times 50 + 28 = 128$ cm (1). | 2+2+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 56 | Half of other diagonal $= \sqrt{35^2 - 21^2} = 28$ |
| 2 | **(d)** 35 | Height difference 21 m; $\sqrt{21^2 + 28^2} = 35$ |
| 3 | **(c)** 294 | Other leg $= 28$; area $= \frac{1}{2}ab$ |
| 4 | **(d)** 21, 72, 75 | Multiples of $(7, 24, 25)$ by 3 stay a triplet |
| 5 | **(d)** 40 | Height splits the base: $h^2 = 50^2 - 30^2 = 1600$ |
| 6 | **(b)** 140 | Other side $= 40$; perimeter $= 2(30 + 40)$ |
| 7 | **(a)** 80 | Half of other diagonal $= \sqrt{50^2 - 30^2} = 40$ |
| 8 | **(b)** 50 | Height difference 30 m; $\sqrt{30^2 + 40^2} = 50$ |
| 9 | **(c)** 600 | Other leg $= 40$; area $= \frac{1}{2}ab$ |
| 10 | **(d)** 39, 52, 65 | Multiples of $(3, 4, 5)$ by 13 stay a triplet |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Half of other diagonal$^2 = 45^2 - 27^2 = 1296$ (1); other diagonal $= 72$ cm (1). | 1+1 |
| 12 | Height bisects the base, $h^2 = 35^2 - 21^2$ (1); $h = 28$ cm (1). | 1+1 |
| 13 | Other side $= 48$ cm (1); area $= 672$ cm² (1). | 1+1 |
| 14 | Half of other diagonal$^2 = 58^2 - 40^2 = 1764$ (1); other diagonal $= 84$ cm (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Initial foot distance $= \sqrt{26^2 - 24^2} = 10$ m (1); new height $= \sqrt{26^2 - 24^2} = 10$ m (1); slip $= 24 - 10 = 14$ m (1). | 1+1+1 |
| 16 | $30^2 + 40^2 = 2500 = 50^2$ (1); right-angled with legs 30, 40 (1); area $= \frac{1}{2} \times 30 \times 40 = 600$ cm² (1). | 1+1+1 |
| 17 | Hypotenuse $= 50$ cm (1); area $= \frac{1}{2} \times 30 \times 40 = 600$ (1); $\frac{1}{2} \times 50 \times h = 600$ gives $h = 24$ cm (1). | 1+1+1 |
| | **OR:** Initial foot distance $= \sqrt{39^2 - 36^2} = 15$ m (1); new height $= \sqrt{39^2 - 36^2} = 15$ m (1); slip $= 36 - 15 = 21$ m (1). | 1+1+1 |
| 18 | $14^2 + 48^2 = 2500 = 50^2$ (1); right-angled with legs 14, 48 (1); area $= \frac{1}{2} \times 14 \times 48 = 336$ cm² (1). | 1+1+1 |
| | **OR:** Hypotenuse $= 50$ cm (1); area $= \frac{1}{2} \times 14 \times 48 = 336$ (1); $\frac{1}{2} \times 50 \times h = 336$ gives $h = 13.44$ cm (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) $\sqrt{30^2 + 40^2} = 50$ m (2); (ii) saved $= 30 + 40 - 50 = 20$ m (1); (iii) total $= 30 + 40 + 50 = 120$ m (2). | 2+1+2 |
| | **OR:** Height$^2 = 45^2 - 27^2 = 1296$, height $= 36$ cm (2); area $= \frac{1}{2} \times 54 \times 36 = 972$ cm² (2); perimeter $= 2 \times 45 + 54 = 144$ cm (1). | 2+2+1 |
| 20 | (i) $\sqrt{14^2 + 48^2} = 50$ m (2); (ii) saved $= 14 + 48 - 50 = 12$ m (1); (iii) total $= 14 + 48 + 50 = 112$ m (2). | 2+1+2 |
| | **OR:** Height$^2 = 58^2 - 40^2 = 1764$, height $= 42$ cm (2); area $= \frac{1}{2} \times 80 \times 42 = 1680$ cm² (2); perimeter $= 2 \times 58 + 80 = 196$ cm (1). | 2+2+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.